

# FocusLens

Real-Time Student Engagement and Attention Monitor

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Midterm Project Proposal – ITAI 1378: Computer Vision  
and AI



# Problem: Lack of Student Engagement

## Solution: FocusLens

- Real-time monitoring of student engagement
- Computer vision detects attention and focus
- Provides instructors with immediate valuable feedback
- Optimized learning environment through data insights



# Solutions - FocusLens

- What will your system do?
  - Monitor a student's attentiveness during online lectures using computer vision on a webcam feed.
- How will it solve the problem?
  - Track facial features and eye closure to classify the student's status as Attentive, Drowsy, or Absent.

Example:

1. Face Detection
  - Reliably locate the user's face in every frame using Object Detection.
2. Facial Landmark Detection
  - Identify and track 6 key points around each eye for the primary engagement metric.
3. Eye Aspect Ratio (EAR) Calculation
  - The core metric: a ratio from eye landmarks. Small EAR = closed eye; large EAR = open eye.
4. Attentiveness Status
  - Use EAR and face presence to classify status into three states: Attentive, Drowsy, or Absent.



# Technical Tools and Models

- Object detection and landmark identification drive the system's core function.
- The proposed solution utilizes Dlib for robust face and landmark detection.
- Core frameworks include Python, OpenCV, NumPy, and Pandas for processing.
- Python is the primary language, standard for computer vision and machine learning.
- Matplotlib or Streamlit will be used for final data visualization and reporting.



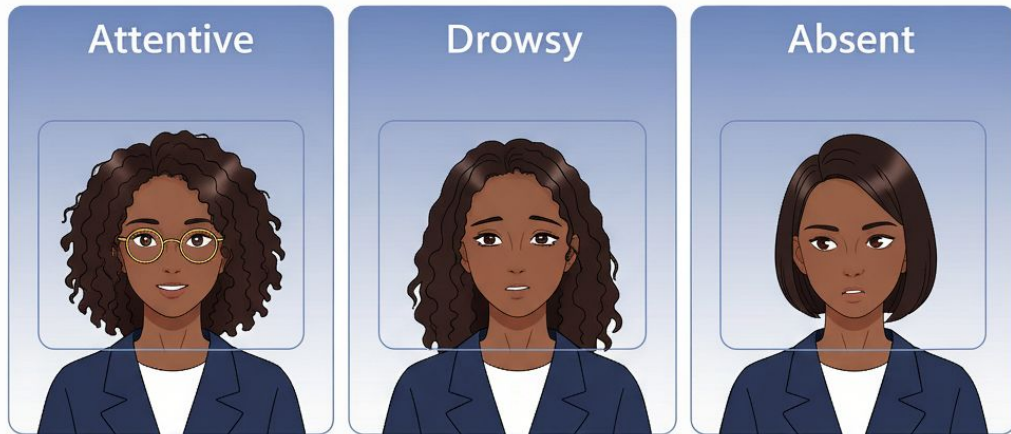
# Data Collection and Plan

- Source a public computer vision dataset for the project.
- The dataset size needs to be determined and justified soon.
- Labels will include Attentive, Drowsy, and Absent classifications.
- Data preparation includes labeling and cleaning the selected images.

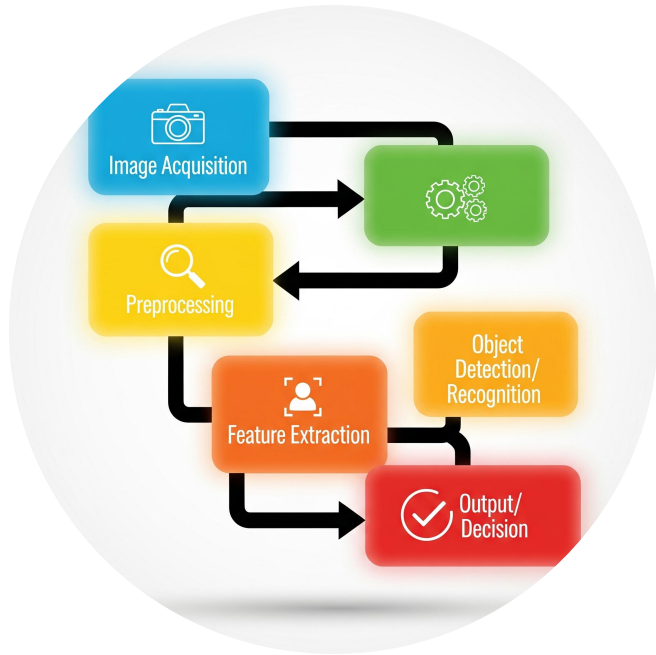


# Project Resources and Tools

- The FocusLens data source can be a public dataset or a new collection
- The required size needs to be carefully determined and fully justified
- Data preparation includes cleaning and labeling the selected images
- Classification labels must include Attentive, Drowsy, and Absent statuses



# System Flow Pipeline



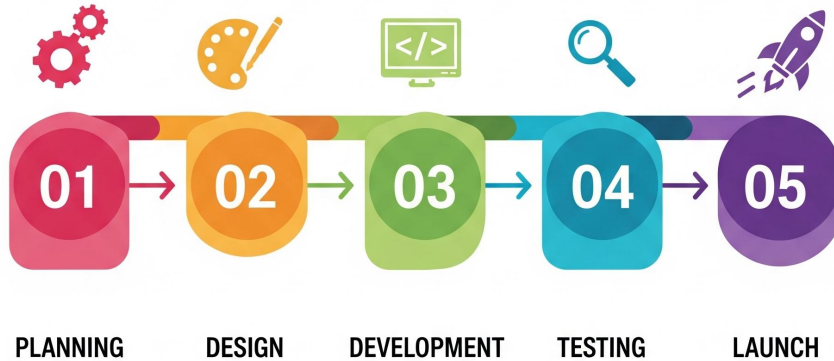
- Webcam acquires video frame input continuously
- Frame undergoes initial processing and object detection
- Facial landmarks and eye regions are precisely analyzed
- System calculates Eye Aspect Ratio for attentiveness status
- Real-time attentiveness status is provided as the final output

# Measuring Project Success

- Achieve high accuracy, precision, and recall scores
- Target a primary metric performance of ninety percent
- Ensure secondary metric speed is less than one second
- Minimize image latency to optimize system performance

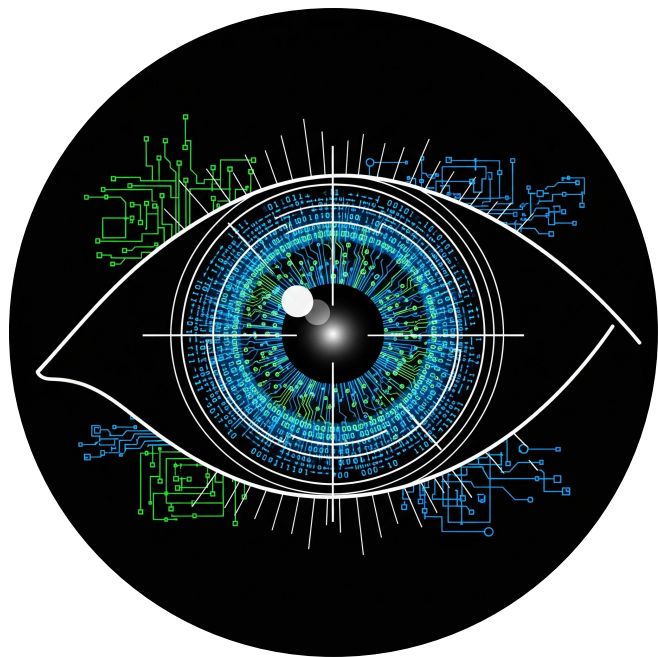


# Project Timeline: 5 Weeks



- Week one - focuses on core system setup and detection
- Week two - Real-time Eye Aspect Ratio calculation is the focus
- Week three - involves implementing the final engagement status logic
- Week four - Data logging and the final report generation are scheduled.
- The last week is dedicated to code cleaning and final submission

# Project Resources: Tools



- FocusLens requires minimal external resources
- Cloud compute options include Colab and Kaggle
- Frameworks like PyTorch can power the models
- The estimated project cost is extremely low